ClipSAT : Multivariable Calculus Partials, Multiple Integrals & Vector Calcu [ANSWER KEY]

Unit 3.01: Multivariable Optimization, Hessians & Lagrange Multipliers

Q1. Consider the function z = fx,y = x^2 + y^2 . Let x = 2t and y = 3t . Use the Multivariable Chain Rule to find frac{dz}{dt} when t=1 .

[ ] A: 10
[ ] B: 13
[x] C: 26
[ ] D: 52

Solution: We are given z = fx,y = x^2 + y^2 , with x = 2t and y = 3t . We need to find frac{dz}{dt} using the Multivariable Chain Rule: frac{dz}{dt} = frac{partial z}{partial x} frac{dx}{dt} + frac{partial z}{partial y} frac{dy}{dt} . First, find the partial derivatives of z with respect to x and y : frac{partial z}{partial x} = 2x frac{partial z}{partial y} = 2y Next, find the derivatives of x and y with respect to t : frac{dx}{dt} = 2 frac{dy}{dt} = 3 Now, substitute these into the chain rule formula: frac{dz}{dt} = 2x2 + 2y3 = 4x + 6y . We are asked to find frac{dz}{dt} when t=1 . First, find the values of x and y when t=1 : x = 21 = 2 y = 31 = 3 Finally, substitute these values of x and y into the expression for frac{dz}{dt} : frac{dz}{dt} bigg|_{t=1} = 42 + 63 = 8 + 18 = 26 .

Q2. Evaluate the double integral of the function fx,y = 2xy over the rectangular region R = [0, 2] times [0, 3] using Fubini's Theorem. Set up the iterated integral by integrating with respect to y first, then x .

[ ] A: 12
[x] B: 18
[ ] C: 6
[ ] D: 24

Solution: To evaluate the double integral iint_R 2xy , dA where R = [0, 2] times [0, 3] , we can use Fubini's Theorem. We will set up the iterated integral by integrating with respect to y first, then x : int_{0}^{2} int_{0}^{3} 2xy , dy , dx First, integrate with respect to y , treating x as a constant: int_{0}^{3} 2xy , dy = left[xy^2 right]_{y=0}^{y=3} = x3^2 - x0^2 = 9x Now, integrate the result with respect to x : int_{0}^{2} 9x , dx = left[frac{9x^2}{2} right]_{x=0}^{x=2} = frac{92^2}{2} - frac{90^2}{2} = frac{94}{2} = frac{36}{2} = 18 Therefore, the value of the double integral is 18.

Q3. For fx,y=7 x^{2} + 4 y^{2} , find f_y-3,1 .

[ ] A: -24
[ ] B: 4
[x] C: 8
[ ] D: -42

Solution: Applying Partial Derivatives principles, f_y=8 y , so f_y-3,1=81=8 . Therefore, the verified correct option is 8.

Q4. Consider the function fx,y = x^2 + y^2 . Find the value of f_{xx}2,3 .

[x] A: 2
[ ] B: 4
[ ] C: 6
[ ] D: 0

Solution: First, we find the first partial derivative of fx,y with respect to x : f_xx,y = frac{partial}{partial x}x^2 + y^2 = 2x . Next, we find the second partial derivative of fx,y with respect to x which is f_{xx} : f_{xx}x,y = frac{partial}{partial x}f_xx,y = frac{partial}{partial x}2x = 2 . Finally, we evaluate f_{xx}x,y at the point 2,3 . Since f_{xx}x,y is a constant, its value at any point is 2. Therefore, f_{xx}2,3 = 2 .

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Q5. Let the linear scalar potential be hu, y = 15u + 8y - 16 . Compute the squared gradient norm |nabla h|^2 .

[x] A: 289
[ ] B: 293
[ ] C: 281
[ ] D: 304

Solution: Since nabla h = langle 15, 8 rangle , |nabla h|^2 = 15^2 + 8^2 = 289 .

Q6. Consider the surface S which is the portion of the plane z = 2x + 3y + 1 lying above the rectangular region R = [0, 1] times [0, 2] in the xy -plane. Calculate the surface area of S .

[ ] A: sqrt{14}
[x] B: 2 sqrt{14}
[ ] C: 3 sqrt{14}
[ ] D: 4 sqrt{14}

Solution: The surface area of a surface z = gx,y above a region R is given by the integral iint_R sqrt{1 + g_x^2 + g_y^2} , dA . Here, gx,y = 2x + 3y + 1 . The partial derivatives are g_x = 2 and g_y = 3 . Therefore, sqrt{1 + g_x^2 + g_y^2} = sqrt{1 + 2^2 + 3^2} = sqrt{1 + 4 + 9} = sqrt{14} . The integral becomes iint_R sqrt{14} , dA , where R = [0, 1] times [0, 2] . This evaluates to sqrt{14} times text{Area}R . The area of the rectangle R is 1-0 times 2-0 = 2 . So, the surface area is 2 sqrt{14} .

Q7. Consider the vector field mathbf{u}s, y, z = langle 22s,, 7y,, 7z rangle on mathbb{R}^3 . Compute the scalar divergence nabla cdot mathbf{u} .

[ ] A: 58
[ ] B: 40
[ ] C: 29
[x] D: 36

Solution: The divergence is 22 + 7 + 7 = 36 .

Q8. Consider the vector field mathbf{F}x,y = langle 3x, 2y rangle . Evaluate the vector field at the point 2, 4 .

[x] A: langle 6, 8 rangle
[ ] B: langle 8, 6 rangle
[ ] C: langle 5, 6 rangle
[ ] D: langle 6, 5 rangle

Solution: The vector field is given by mathbf{F}x,y = langle 3x, 2y rangle . To evaluate the vector field at the point 2, 4 , we substitute x=2 and y=4 into the components of the vector field. First component: 3x = 32 = 6 . Second component: 2y = 24 = 8 . Therefore, mathbf{F}2,4 = langle 6, 8 rangle .